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Implicit guarantees and the rise of shadow banking: The case of trust products

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1 Big picture

This paper studies *implicit guarantees* in China’s trust-product market and argues that these guarantees are a key mechanism behind the rise (and later contraction) of shadow banking. The central idea is that trust products are *debt-like* claims sold to retail/wealth-management investors, but investors are partially protected by intermediaries (trust firms and affiliated institutions) through *implicit* (non-contractual) guarantees that are honored in “normal” states but can fail in aggregate bad states. Because of these guarantees, promised yields/spreads do *not* only price underlying borrower risk; they also price the *strength of the implicit guarantee*. Empirically, the paper shows (i) yield spreads are lower when guarantee proxies are stronger, (ii) yield spreads respond more to risk when guarantees are weaker, and (iii) 2018 regulations restricting implicit guarantees weaken the yield–guarantee relationship and lead to greater credit rationing for non-SOEs, consistent with worsened capital misallocation.

2 Institutional setting and data (what is being measured)

- **Asset:** Chinese trust products (a major shadow-banking channel). Products raise funds from investors and invest in a specific borrower/project. Prospectuses disclose product terms (promised yield, maturity, structure, collateral, investment threshold) and the investment target’s characteristics (industry, location, some size information).
- **Key outcome: expected yield spread** (in % points) defined as the promised yield minus a maturity-matched Treasury yield (the paper treats the risk-free rate as normalized in the model; empirically, “spread” is the relevant object).
- **Main “implicit guarantee” proxies:**
 1. **Central SOE:** trust company controlled by the central government;
 2. **Local SOE:** trust company controlled by local government;
 3. **Registered capital:** size/financial strength of trust firm (often used as a capacity-to-guarantee proxy);
 4. **Sale bank big5:** product distributed via one of the Big Five state banks;
 5. **Large trust firm/shareholder size:** additional balance-sheet strength proxy.
- **IG index:** an additive index that aggregates multiple guarantee proxies into a discrete measure of “guarantee strength.”
- **Risk proxies (underlying borrower/project risk):**
 1. **i.real estate:** indicator that the investment target is real estate (higher systematic/sector risk in the trust market);
 2. **GDP growth:** city/province GDP growth where borrower is located (treated as *inverse* risk: lower growth $\Rightarrow$ riskier);

3. **Borrower size:** proxy for borrower scale (treated as *inverse* risk: smaller firms are riskier);
4. **Housing market risk:** additional risk proxy within real estate products.

3 Model 1: a screening-and-guarantee contracting model (core theory)

3.1 Environment

An intermediary (trust firm) channels investors' funds into projects. Each project requires 1 unit of investment. Project payoff is symmetric around 1: success payoff $1 + \delta$, failure payoff $1 - \delta$. A project can be *good* or *bad*. Bad projects always fail. Good projects fail with probability p and succeed with probability $1 - p$. Let q be the probability that a randomly drawn project is bad.

The intermediary can **screen** (effort $e \in [0, 1]$) to reduce exposure to bad projects. Screening cost is linear: ξe . The paper uses the key ratio

$$\eta \equiv \frac{\xi}{q\delta}, \quad (1)$$

which is the screening cost scaled by the expected loss from a bad project.

3.2 Implicit guarantee mechanism

A defining feature is that if a project fails, the intermediary may (informally) *compensate* the investor to honor the promised yield, but doing so requires costly external finance. The unit cost of raising funds is state-dependent:

$$\phi = \begin{cases} \phi_N, & \text{normal state,} \\ \phi_B, & \text{bad aggregate state,} \end{cases} \quad \phi_B > \phi_N \geq 1.$$

Let π be the probability of a bad aggregate state conditional on failure. A higher π means a weaker effective guarantee (because the guarantee is less likely to be honored).

3.3 Intermediary objective under general contracts

Let D_b be the intermediary's net payoff (after the investor is paid) in a given state. If $D_b < 0$, the intermediary must raise $-D_b$ externally at unit cost ϕ , so the extra financing wedge is $(\phi - 1)(-D_b)$. The objective can be written as

$$\max_{(\text{contract}, e)} \mathbb{E}[D_b] + \mathbb{E}[(\phi - 1) \min(D_b, 0)] - \xi e, \quad (2)$$

subject to the investor's participation constraint (risk-neutral investor with normalized required return) and feasibility.

3.4 Proposition 1 (first-best vs second-best, unrestricted contract space)

First-best:

- If $\eta < 1$, screening is efficient ($e = 1$) and the intermediary's utility is

$$u_{1b} = u_{ns} + q\delta(1 - \eta),$$

where u_{ns} is the no-screening benchmark.

- If $\eta \geq 1$, the intermediary does not screen and obtains u_{ns} .

Second-best with costly external finance: there is a threshold $\eta_{2b} < 1$ such that if $\eta < \eta_{2b}$ the intermediary screens and uses a *conditional guarantee*: it compensates investors in the normal state but not in the bad state. If $\eta \geq \eta_{2b}$, it neither screens nor guarantees. The paper emphasizes the moral-hazard wedge:

$$u_{1b} - u_{2b} = (\rho_{2b} - 1)\xi > 0,$$

where $\rho_{2b} > 1$ reflects the distortion induced by costly external finance and limited commitment.

3.5 Debt-like payoff restriction and Proposition 2 (four cases)

The paper then restricts investor payoffs to mimic debt contracts:

- if the project succeeds: investor receives $1 + y$,
- if the project fails: investor receives the project payoff $1 - \delta$,
- the intermediary can choose whether to make up the shortfall $y + \delta$ via an (implicit) guarantee.

Under this restriction, the optimal contract has four regions in η (screening cost) with cutoffs $\eta_0 < \eta_1 < \eta_{ug}$:

Case 1 $\eta \leq \eta_0$: screen and choose a **conditional guarantee**. The yield is

$$y = \frac{p\pi\delta}{1 - p\pi}, \tag{3}$$

investor utility is 0, and intermediary utility is

$$u_{cg} = u_{ns} + q\delta(\eta_{cg} - \eta).$$

Case 2 $\eta \in (\eta_0, \eta_1)$: screen and choose a **conditional guarantee**, but now the investor obtains *positive rent* and the yield is pinned down by incentive considerations:

$$y = \left(\frac{\eta}{(1 - \pi)\phi_N} - 1 \right) \delta. \tag{4}$$

Case 3 $\eta \in [\eta_1, \eta_{ug}]$: screen and choose an **unconditional guarantee**. The yield is $y = 0$ and the intermediary utility is

$$u_{ug} = u_{ns} + q\delta(\eta_{ug} - \eta).$$

Case 4 $\eta > \eta_{ug}$: **no screening** and **no guarantee**. The yield is

$$y = \frac{(1-q)p + q}{(1-q)(1-p)}. \quad (5)$$

(This expression is the “failure odds” under no screening; under the paper’s payoff normalization, it is the required yield to satisfy the risk-neutral investor’s participation constraint.)

3.6 Proposition 3: comparative statics under conditional guarantees

When conditional guarantees are optimal (roughly $\eta \leq \eta_1$), the yield (yield spread) depends on two primitives:

- p : underlying project risk (higher p means more likely failure even for good projects),
- π : probability that the economy is in a bad state when failure occurs (a *guarantee-strength* parameter).

The paper shows (writing the safe rate as r in the theory):

$$\frac{\partial y}{\partial p} = \begin{cases} \frac{\pi(\delta + r)}{(1 - p\pi)^2} > 0, & \eta \leq \eta_0, \\ 0, & \eta_0 < \eta \leq \eta_1, \end{cases} \quad (6)$$

$$\frac{\partial y}{\partial \pi} = \begin{cases} \frac{p(\delta + r)}{(1 - p\pi)^2} > 0, & \eta \leq \eta_0, \\ \frac{\eta}{(1 - \pi)^2 \phi_N} - \delta > 0, & \eta_0 < \eta \leq \eta_1, \end{cases} \quad (7)$$

$$\frac{\partial^2 y}{\partial \pi \partial p} = \begin{cases} \frac{(1 + p\pi)(\delta + r)}{(1 - p\pi)^3} > 0, & \eta \leq \eta_0, \\ 0, & \eta_0 < \eta \leq \eta_1. \end{cases} \quad (8)$$

Economic mapping to the empirics:

1. higher underlying risk (p) raises yield spreads;
2. higher bad-state probability (π) raises yield spreads because the guarantee is less likely to be honored;
3. stronger guarantees imply *lower spread levels* and *lower spread sensitivity* to risk.

3.7 Repeated game / reputation foundation for implicit guarantees

The paper also provides a reputation-based foundation (in a repeated-game setting) for why guarantees are honored in normal times but not in aggregate bad states. A key sufficient condition is

that the gap in external-finance costs is large enough:

$$\phi_B - \phi_N \geq \hat{\phi}, \quad (9)$$

where $\hat{\phi}$ is a threshold (given in the Online Appendix). Intuition: if the intermediary fails to honor the guarantee in a normal state, investors infer that it is in a much worse state, causing a reputational penalty large enough that honoring in normal times is incentive compatible.

4 Model 2: implicit guarantees and capital misallocation (SOE vs private)

The paper embeds the above contract logic into a two-sector allocation framework. Let p index projects, with p interpreted as an *inverse* productivity measure (lower p implies higher productivity). Projects exist in SOE and private sectors and are drawn from distribution $F(\cdot)$. Total capital supply is fixed at 1. An equilibrium is characterized by sector-specific funding cutoffs p_{SOE} and p_{PS} and an equilibrium interest rate r .

4.1 Socially optimal benchmark

Assuming the same project distribution in the SOE and private sectors, the social planner funds all projects with $p \in [0, p_{\text{social}}]$ in *both* sectors, where

$$p_{\text{social}} = F^{-1}\left(\frac{1}{2(1-q)}\right). \quad (10)$$

Total output under the social optimum is

$$\Pi_{\text{social}} = 1 + \delta - 4(1-q)\delta \cdot \mathbb{E}(p \mid p \leq p_{\text{social}}). \quad (11)$$

4.2 Traditional banking (TB) regime

In TB, the intermediary provides bank loans with (effectively) explicit guarantees. In equilibrium, projects with $p \in [0, p_{SOE, TB}]$ (SOE) and $p \in [0, p_{PS, TB}]$ (private) are funded. The triple $(p_{SOE, TB}, p_{PS, TB}, r_{TB})$ is pinned down by:

$$p_{SOE, TB} = \frac{1}{2} - \frac{r_{TB}}{2\delta} - \frac{\xi}{2\delta(1-q)}, \quad (12)$$

$$p_{PS, TB} = \left(1 - \frac{(\bar{\phi} - 1)(r_{TB} + \delta)}{2\delta + (\bar{\phi} - 1)(r_{TB} + \delta)}\right) p_{SOE}(r_{TB}), \quad (13)$$

$$(1-q)[F(p_{SOE, TB}) + F(p_{PS, TB})] = 1. \quad (14)$$

Total output in TB is

$$\Pi_{TB} = 1 + \delta - 2(1-q)\delta \left[\mathbb{E}(p \mid p \leq p_{SOE, TB}) + \mathbb{E}(p \mid p \leq p_{PS, TB}) \right]. \quad (15)$$

4.3 Extended banking (EB) regime with shadow banking

In EB, the intermediary still uses bank loans (with explicit guarantees) to fund SOE projects, but uses shadow banking with *implicit guarantees* to fund private projects. The equilibrium $(p_{SOE,EB}, p_{PS,EB}, r_{EB})$ solves:

$$p_{SOE,EB} = \frac{1}{2} - \frac{r_{EB}}{2\delta} - \frac{\xi}{2\delta(1-q)}, \quad (16)$$

$$(1-q) \left[(1-2p_{PS,EB})\delta - r_{EB} - p_{PS,EB}(1-\pi)(\phi_N - 1)(y + \delta) \right] = \xi, \quad (17)$$

$$(1-q) [F(p_{SOE,EB}) + F(p_{PS,EB})] = 1, \quad (18)$$

where the loan yield y follows the contract solution (it depends on whether the relevant η cutoff binds; see Proposition 2). Total output in EB is

$$\Pi_{EB} = 1 + \delta - 2(1-q)\delta \left[\mathbb{E}(p \mid p \leq p_{SOE,EB}) + \mathbb{E}(p \mid p \leq p_{PS,EB}) \right]. \quad (19)$$

4.4 Proposition 4 and corollaries (interpretation)

The paper's comparative statics imply:

- **Misallocation in both regimes:** $p_{SOE} > p_{social} > p_{PS}$ (too many SOE projects and too few private projects funded).
- **Shadow banking mitigates misallocation:** $p_{SOE,EB} < p_{SOE,TB}$ and $p_{PS,EB} > p_{PS,TB}$, so capital shifts from SOEs to private sector in EB.
- **Output ranking:** $\Pi_{TB} < \Pi_{EB} < \Pi_{social}$.
- **Regulation that raises the cost of implicit guarantees** (modeled as a cost c if an intermediary uses an implicit guarantee) reduces private funding and worsens misallocation; formally, the paper states $dp_{PS,EB}/dc > 0$, $d(p_{SOE,EB} - p_{PS,EB})/dc > 0$, and $d\Pi_{EB}/dc < 0$.

5 Empirical models (what is estimated)

5.1 Baseline cross-sectional pricing regression (Table 3)

The core empirical specification maps the theory's "yield depends on risk and guarantee strength" into:

$$\text{YieldSpread}_i = \alpha + \beta \cdot \text{IGStrength}_i + \gamma \cdot \text{Risk}_i + \Gamma' X_i + \text{FE} + \varepsilon_i, \quad (20)$$

where i indexes trust products, IGStrength is proxied by (Central SOE, Local SOE, log registered capital, sale bank big5, shareholder size) or summarized by the IG index, Risk is proxied by (real-estate indicator, GDP growth, borrower size), and X_i includes product controls (maturity, structured product indicator, investment threshold, collateral).

5.2 Shock to guarantee strength: 2015 stock market crash (Table 4, Fig. 3)

To isolate changes in perceived guarantee strength, the paper uses the 2015 stock market crash and measures each trust firm’s **stock exposure** (pre-crash). The reduced-form DiD is:

$$\text{YieldSpread}_{i,t} = \alpha + \beta_1(\text{StkExposure}_i \times \text{PostCrash}_t) + \beta_2\text{StkExposure}_i + \beta_3\text{PostCrash}_t + \Gamma'X_{i,t} + \text{FE} + \varepsilon_{i,t}. \quad (21)$$

Theory sign: if the crash weakens the intermediary’s capacity to honor guarantees, then yields should rise more for higher-exposure trust firms: $\beta_1 > 0$.

5.3 Risk-sensitivity test (Table 5)

A direct implication of the model is that strong guarantees dampen the sensitivity of yields to risk. The paper estimates:

$$\text{YieldSpread}_i = \alpha + \beta \cdot \text{IGStrength}_i + \gamma \cdot \text{Risk}_i + \theta \cdot (\text{IGStrength}_i \times \text{Risk}_i) + \Gamma'X_i + \text{FE} + \varepsilon_i. \quad (22)$$

Theory sign: θ should reduce the magnitude of risk pricing. For a risk proxy where higher value means higher risk (e.g., `i.real estate`), the paper expects $\theta < 0$.

5.4 Shocks to underlying risk (Tables 6–8, Figs. 4 and 6)

The paper uses two events that shift perceived underlying risk:

1. **First high-profile default (Jan. 2014):** increases perceived trust-product risk.
2. **Housing purchase restrictions (Order 10, Apr. 2010):** increases risk for real-estate related trust lending in affected provinces.

Both are implemented with DiD-type designs:

$$\text{YieldSpread}_{i,t} = \alpha + \beta_1\text{PostShock}_t + \beta_2(\text{PostShock}_t \times \text{IGStrength}_i) + \beta_3\text{IGStrength}_i + \Gamma'X_{i,t} + \text{FE} + \varepsilon_{i,t}, \quad (23)$$

often complemented with event-time interactions to visualize pre-trends and dynamic effects.

5.5 Policy shock: 2018 regulations restricting implicit guarantees (Tables 10–11, Figs. 7–8)

The 2018 asset-management regulations are treated as a structural break in the implicit-guarantee regime. A pricing regression is:

$$\text{YieldSpread}_{i,t} = \alpha + \beta_1(\text{PostReg}_t \times \text{IGStrength}_{i,t}) + \beta_2\text{IGStrength}_{i,t} + \Gamma'X_{i,t} + \text{FE} + \varepsilon_{i,t}. \quad (24)$$

Prediction: if implicit guarantees are curtailed, the spread advantage of high-IG products shrinks, i.e., $\beta_1 > 0$ when $\beta_2 < 0$.

A quantity/credit-rationing regression studies issuance size and non-SOE borrowers:

$$\log(\text{LoanSize}_{i,t}) = \alpha + \beta_1(\text{NonSOE}_i \times \text{PostReg}_t) + \beta_2(\text{IGStrength}_{i,t} \times \text{NonSOE}_i \times \text{PostReg}_t) + \Gamma' X_{i,t} + \text{FE} + \varepsilon_{i,t}. \quad (25)$$

Prediction: restrictions on guarantees reduce credit to riskier non-SOEs: $\beta_1 < 0$, and stronger-IG intermediaries attenuate the contraction: $\beta_2 > 0$.

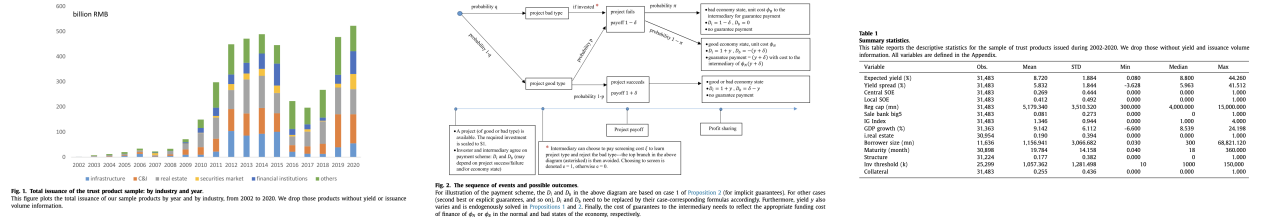
6 All tables and figures

6.1 Fig. 1: time-series issuance patterns

Fig. 1 plots total issuance in the sample by industry over 2002–2020. The figure documents the rapid expansion of trust-product financing after the late 2000s, and highlights real estate and C&I as major destinations. This motivates the paper’s focus on how trust products grew into a key shadow-banking channel.

6.2 Fig. 2: sequence of events in the model

Fig. 2 is the model timeline illustrating why a guarantee can be *implicit* (state-contingent): after a failure, the intermediary can fully compensate investors in the normal state (at cost ϕ_N) but not in the bad state (high cost ϕ_B). This microfoundation links yields to both project risk (p) and guarantee strength (π).



6.3 Table 1: summary statistics

Key numbers to remember:

- Mean expected yield: 8.720%; mean yield spread: 5.832% (large pricing wedge relative to Treasuries).
- Central SOE trust firms: 26.9% of products; Local SOE: 41.2%.
- Sale bank big5 is relatively rare in this sample (8.1%), but is interpreted as a strong distribution-based guarantee proxy.
- IG index has mean 1.346 (median 1; range 0–4), indicating substantial heterogeneity in implied guarantee strength.

6.4 Table 2: industry distribution

Real estate accounts for about 23.9% of issuance volume, C&I for 19.4%, infrastructure for 14.3%, and financial institutions for 11.3%. This composition is consistent with the view that trust products fund credit segments that are not fully served by traditional bank lending.

Table 2
Industry distribution of investment capital raised from trust products: 2002-2020

This table reports the industry distribution of investment capital raised through trust product issuance in our sample from 2002 to 2020.

Industries	Total issuance volume (in \$Bn)	Percentage (%)	Product #	Percentage of product # (%)
Real estate	1303.55	23.92	5,947	18.80
C&I	815.27	19.44	6,259	19.88
Infrastructure	598.76	14.27	5,487	18.06
Financial institutions	474.34	11.31	2,998	9.52
Securities market	215.50	5.16	2,462	8.00
Others	1986.37	25.90	7,790	24.74
TOTAL	4,194.78	100	31,483	100

Table 3
Determinants of yield spreads: The role of implicit guarantees and underlying risks

This table reports the results from examining the determinants of on-site yield spreads (the difference between expected yield at issue and the realized (tenure-based) interest rate) based on the events of issuance and maturity of the trust products. The dependent variable is the product yield spread. Panel A reports the regression results using trust firm characteristics such as ownership type (Central SOE, Local SOE, non-firm size (log big reg)), and the size of the trust firms controlling standardized log (total size) as the trust explanatory variables. Panel B reports the results using the IG index as the trust explanatory variable. IG index is calculated as the sum of the values of SOE, sale bank big5, and large firm indicators, where SOE equals to 1 for central SOE, 0.5 for local SOE, and 0 for non-SOE; sale bank big5 equals to 1 if the product is sold by a sale bank, 0.5 if the product is issued by a trust firm whose registered capital is in the top five among all trust firms, and 0 otherwise; large firm indicator is defined as 1 if the product is issued by a trust firm whose registered capital is in the top five among all trust firms. All other variables are defined as in the Appendix. Standard errors are clustered at the firm and year levels and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5% and 10% level, respectively.

Panel A: Implicit guarantee and on-site yield spreads

	(1)	(2)	(3)	(4)	(5)	(6)
Implicit guarantee variables						
Central SOE	-0.730***	-0.489***	-0.567***	-0.454***		
	(0.111)	(0.127)	(0.123)	(0.127)		
Local SOE	-0.390***	-0.387***	-0.372***	-0.463***		
	(0.060)	(0.061)	(0.121)	(0.107)		
Log big reg	-0.240***	-0.182***	-0.439***	-0.215***		
	(0.026)	(0.026)	(0.062)	(0.070)		
Sale bank big5	-0.272***	-0.139**	-0.144	-0.0474	-0.202***	-0.0674
	(0.078)	(0.061)	(0.091)	(0.075)	(0.078)	(0.091)
Log total size			-0.0440	-0.0894**		
			(0.014)	(0.020)		
Business characteristics						
Line estate	0.270***	0.323***	0.147*	0.231***	0.190***	0.275***
	(0.054)	(0.061)	(0.074)	(0.062)	(0.051)	(0.056)
GDP growth	-0.027***	-0.027***	-0.014	-0.00362	-0.00747*	
	(0.00862)	(0.00862)	(0.010)	(0.00862)	(0.00862)	
Log borrower size	-0.00151	-0.00151	-0.00151	-0.00151	-0.00151	
	(0.00151)	(0.00151)	(0.00151)	(0.00151)	(0.00151)	
Leadsurance	-0.0789	-0.081	0.0019	-0.00111	-0.00830	-0.0783
	(0.0573)	(0.0573)	(0.051)	(0.0573)	(0.0573)	(0.0573)
Leadsurance market	-1.358***	-0.444*	-1.131***	-0.258	-1.381***	-0.387*
	(0.260)	(0.260)	(0.260)	(0.260)	(0.260)	(0.260)
Leadsurance market	-0.454*	-0.368**	-0.417*	-0.376*	-0.376*	-0.395*
	(0.120)	(0.120)	(0.120)	(0.120)	(0.120)	(0.120)
Leadsurance market	-0.389**	-0.344*	-0.371*	-0.376*	-0.376*	-0.376*
	(0.167)	(0.167)	(0.167)	(0.167)	(0.167)	(0.167)
Product characteristics						
Maturity	0.0222***	0.0495***	0.0279***	0.0462***	0.0222***	0.0462***
	(0.00146)	(0.00146)	(0.00146)	(0.00146)	(0.00146)	(0.00146)
Structure	0.208***	0.348***	0.362***	0.335***	0.280***	0.273***
	(0.0805)	(0.0775)	(0.171)	(0.0824)	(0.0768)	(0.0768)
Log loss threshold	0.12***	0.109**	0.104*	0.100	0.100*	0.100*
	(0.036)	(0.036)	(0.171)	(0.036)	(0.036)	(0.036)
Collateral	0.380***	0.191***	0.344***	0.120*	0.467***	0.209**
	(0.0676)	(0.0676)	(0.0676)	(0.0676)	(0.0676)	(0.0676)
Cores	1.36***	1.36***	1.36***	1.36***	1.36***	1.36***
	(0.406)	(0.406)	(0.406)	(0.406)	(0.406)	(0.406)
Year FE	YES	YES	YES	YES	YES	YES
Trust firm FE	NO	NO	NO	NO	YES	YES
Province FE	YES	YES	YES	YES	YES	YES
Obs.	24,292	10,298	12,051	9,810	24,298	10,295
Adj. R-sq.	0.465	0.475	0.483	0.470	0.517	

(continued on next page)

Table 3
(continued)

Panel B: Implicit guarantee and on-site yield spreads: IG index

	(1)	(2)	(3)	(4)
IG index	-0.354***	-0.269***	-0.269***	-0.286***
	(0.0491)	(0.0491)	(0.0491)	(0.0491)
Business characteristics				
Line estate	0.240***	0.310***	0.190***	0.291***
	(0.052)	(0.052)	(0.052)	(0.052)
GDP growth	-0.027***	-0.027***	-0.027***	-0.027***
	(0.00862)	(0.00862)	(0.00862)	(0.00862)
Log borrower size	-0.00151	-0.00151	-0.00151	-0.00151
	(0.00151)	(0.00151)	(0.00151)	(0.00151)
Leadsurance	-0.0789	-0.0789	-0.0789	-0.0789
	(0.0573)	(0.0573)	(0.0573)	(0.0573)
Leadsurance market	-1.358***	-1.358***	-1.358***	-1.358***
	(0.260)	(0.260)	(0.260)	(0.260)
Leadsurance market	-0.454*	-0.454*	-0.454*	-0.454*
	(0.120)	(0.120)	(0.120)	(0.120)
Leadsurance market	-0.389**	-0.389**	-0.389**	-0.389**
	(0.167)	(0.167)	(0.167)	(0.167)
Product characteristics				
Maturity	0.0222***	0.0495***	0.0222***	0.0495***
	(0.00146)	(0.00146)	(0.00146)	(0.00146)
Structure	0.208***	0.348***	0.208***	0.348***
	(0.0805)	(0.0805)	(0.0805)	(0.0805)
Log loss threshold	0.12***	0.109**	0.12***	0.109**
	(0.036)	(0.036)	(0.036)	(0.036)
Collateral	0.380***	0.191***	0.380***	0.191***
	(0.0676)	(0.0676)	(0.0676)	(0.0676)
Cores	1.36***	1.36***	1.36***	1.36***
	(0.406)	(0.406)	(0.406)	(0.406)
Year FE	YES	YES	YES	YES
Trust firm FE	NO	NO	NO	NO
Province FE	YES	YES	YES	YES
Obs.	24,292	10,298	24,292	10,295
Adj. R-sq.	0.462	0.474	0.469	0.514

6.5 Table 3: baseline pricing evidence (Hypothesis I)

Panel A (individual guarantee proxies). Across specifications, stronger implicit-guarantee proxies are associated with *lower* yield spreads:

- Central SOE: coefficients around -0.45 to -0.71 (statistically significant), indicating central SOE-controlled trust firms issue products with meaningfully lower spreads.
- Local SOE: also negative (around -0.36 to -0.46).
- Higher registered capital: negative and significant (e.g., around -0.18 to -0.44).
- Sale bank big5: negative (e.g., -0.27 in a baseline column), consistent with investors valuing distribution via state megabanks as a guarantee signal.

Risk proxies behave as expected:

- `i.real estate` enters positively (riskier sector $\Rightarrow$ higher spread),
- GDP growth enters negatively (higher growth $\Rightarrow$ safer borrower environment),
- borrower size enters negatively when available (larger borrowers $\Rightarrow$ lower spread).

Panel B (IG index). The IG index is strongly negatively related to yield spreads. A representative coefficient is about -0.354 (s.e. ≈ 0.040), implying that moving one unit up in the IG index lowers spreads by about 0.35 percentage points, holding risk and product characteristics fixed.

6.6 Table 4 and Fig. 3: stock-crash evidence that guarantees matter

Table 4 implements the 2015 stock market crash test. The interaction $\text{StkExposure} \times \text{PostCrash}$ is positive and statistically significant (about 0.072 in the simplest specification), meaning:

after the crash, trust firms with higher stock exposure (weaker balance sheets / weaker guarantee capacity) issue products with larger increases in yield spreads.

Fig. 3 shows the event-time coefficients: pre-trends are small, while post-crash coefficients become positive, supporting the interpretation as a *guarantee-strength* shock rather than differential pre-trends.

Table 4
Stock market crash and yield spreads of trust products.

This table reports the results of regressions examining the impact of stock market crash on trust products' ex ante pricing. The dependent variable is the product yield spread. *Post crash* is equal to 1 if the product was issued between July 1, 2015 to December 31, 2016 (in the 18 months after the stock market crash), and 0 if the product was issued between December 1, 2013 to May 31, 2015 (in the 18 months before the stock market crash). Indicator variables *Time(t)* are defined for each 6-month period around the market crash in the 3-year period, e.g., *Time(0)* equals to 1 if the product was issued during July 2015 – December 2015, the first six months after the crash; *Time(-1)* equals to 1 if the product issued during December 2014 – May 2015, the six-month immediately before the crash, and so forth. *Stk exposure* is defined as the natural logarithm of the volume of the securities market products that was outstanding by the end of May 2015 and due between July to December in 2015. Other controls include *Maturity*, *Structure*, *Long Inv Threshold*, *Collateral*, *IG index*, *GDP growth*, *Real estate*, *Infrastructure*, *Securities market*, *Life institutions*, and *Lothens*. All other variables are defined in the Appendix. Standard errors are clustered at the firm and year levels and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5% and 10% level, respectively.

	Dep. Var. = Product expected yield spread (Y)	
	(1)	(2)
Stk exposure	-0.00246 (0.0206)	-0.0179 (0.0291)
Post crash	-1.260*** (0.342)	
Stk exposure × Post crash	0.0719*** (0.0348)	
		0.0478 (0.0342)
Stk exposure × Time(-3)		0.0208 (0.0384)
Stk exposure × Time(-2)		0.00189 (0.0355)
Stk exposure × Time(-1)		0.0932** (0.0410)
Stk exposure × Time(1)		0.0772* (0.0374)
Cons.	YES	YES
Other controls	YES	YES
Time FE	NO	YES
Obs	6,802	6,888
adj. R-sq	0.465	0.517

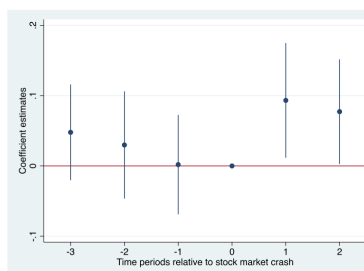


Fig. 3. Stock market crash and yield spreads of trust products.
This figure plots the time trend of the treatment effect estimates around the stock market crash in June 2015, as explained in Section 5.1 and in Column 2 of Table 4. For each time period, we plot the point estimate (the solid circle) and the 95% confidence interval (the vertical lines intersecting the solid circles). Time indicators are defined for each 6-month period around the market crash in the 3-year period, e.g., *Time(0)* denotes the period during July 2015 – December 2015, the first six months after the crash; *Time(-1)* denotes the period during December 2014 – May 2015, the six-month immediately before the crash, and so forth.

Table 5
Spread sensitivity to the investment risks of trust products: the role of explicit guarantees.

This table reports the results of regressions examining the effects of inventory exposures of implicit guarantees on the risk sensitivity of product pricing. The dependent variable is the product expected yield spread. The key explanatory variable, *IG index*, is defined as the ratio of the product's expected yield spread to the expected yield spread of the product with the highest IG index. *Stk exposure* is defined as the natural logarithm of the volume of the securities market products that was outstanding by the end of May 2015 and due between July to December in 2015. Other controls include *Maturity*, *Structure*, *Long Inv Threshold*, *Collateral*, *IG index*, *GDP growth*, *Real estate*, *Infrastructure*, *Securities market*, *Life institutions*, and *Lothens*. All other variables are defined in the Appendix. Standard errors are clustered at the firm and year levels and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5% and 10% level, respectively.

	Dep. Var. = Product expected yield spread (Y)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
GDP growth	-0.0088*** (0.00142)	-0.0177*** (0.00777)	-0.0254*** (0.0102)	-0.0181*** (0.00983)	-0.0288*** (0.00986)	-0.0268*** (0.00972)	-0.0248*** (0.00977)	-0.0146*** (0.00991)
Real estate	0.158 (0.0963)	0.499** (0.1742)	0.791*** (0.1606)	0.467*** (0.1611)	0.271*** (0.0886)	0.480*** (0.0861)	0.264*** (0.0944)	0.336*** (0.0839)
Log borrower size		-0.0001 (0.0184)		-0.0001 (0.0143)		-0.0001 (0.0125)		-0.0027* (0.0133)
IG index		-0.481*** (0.180)		-0.485*** (0.180)		-0.485*** (0.180)		-0.485*** (0.180)
IG index × GDP growth		0.0028 (0.00632)		0.00172 (0.00648)		0.00172 (0.00648)		0.00172 (0.00648)
IG index × Real estate		-0.0012 (0.0099)		-0.0012 (0.0099)		-0.0012 (0.0099)		-0.0012 (0.0099)
IG index × Log borrower size		-0.0009 (0.0119)		-0.0009 (0.0119)		-0.0009 (0.0119)		-0.0009 (0.0119)
Control SIC		-0.0001 (0.180)		-0.0001 (0.180)		-0.0001 (0.180)		-0.0001 (0.180)
Control SIC × GDP growth		0.0001 (0.0111)		0.0001 (0.0111)		0.0001 (0.0111)		0.0001 (0.0111)
Control SIC × Real estate		-0.0001 (0.128)		-0.0001 (0.128)		-0.0001 (0.128)		-0.0001 (0.128)
Control SIC × Log borrower size		-0.0001 (0.0119)		-0.0001 (0.0119)		-0.0001 (0.0119)		-0.0001 (0.0119)
Large effect		-0.0001 (0.172)		-0.0001 (0.172)		-0.0001 (0.172)		-0.0001 (0.172)
Large effect × GDP growth		0.0001 (0.0111)		0.0001 (0.0111)		0.0001 (0.0111)		0.0001 (0.0111)
Large effect × Real estate		-0.0001 (0.128)		-0.0001 (0.128)		-0.0001 (0.128)		-0.0001 (0.128)
Large effect × Log borrower size		-0.0001 (0.0119)		-0.0001 (0.0119)		-0.0001 (0.0119)		-0.0001 (0.0119)
Safe bank high								
Safe bank high × GDP growth								
Safe bank high × Real estate								
Safe bank high × Log borrower size								
Cons.	YES	YES	YES	YES	YES	YES	YES	YES
Other controls	YES	YES	YES	YES	YES	YES	YES	YES
Time FE	YES	YES	YES	YES	YES	YES	YES	YES
Province FE	YES	YES	YES	YES	YES	YES	YES	YES
adj. R-sq	0.464	0.475	0.456	0.464	0.441	0.462	0.431	0.449

6.7 Table 5: guarantees dampen risk pricing (Hypothesis II)

Table 5 estimates interaction models and shows that stronger guarantees reduce spread sensitivity to risk proxies:

- For real estate risk, the interaction between IG measures and *real estate* is typically *negative* (e.g., $\text{IG index} \times \text{real estate}$ is negative and statistically significant in the richer specification), meaning high-IG products price the real-estate premium less strongly.
- For inverse-risk proxies (GDP growth and borrower size), interactions tend to move the coefficient toward zero (i.e., attenuate absolute risk sensitivity), consistent with the theory that guarantees absorb part of downside risk.

6.8 Table 6 and Fig. 4: default shock (risk increases) and heterogeneous pricing

After the first high-profile default (Jan. 2014), spreads increase on average (PostDefault coefficient ≈ 0.387 in the simplest specification). Crucially, the interaction $\text{PostDefault} \times \text{IG index}$ is negative (about -0.241), implying:

high-IG products experience a smaller spread increase after a rise in perceived underlying risk.

Fig. 4 visualizes this pattern in event time: post-default interaction coefficients turn negative, while pre-default coefficients are comparatively flat.

Table 6
The first high-profile default case as a negative shock to perceived underlying risk.

This table reports the results of regressions examining the impact of the first high-profile default case in the trust industry (in January 2014) on the pricing of implicit guarantees. The dependent variable is the product yield spread. *Post default* is defined as 1 if the product is issued after February 1, 2014, and 0 otherwise. We consider the sample of products issued in the 24 months before and after the default case, i.e., February 2012 to January 2016. Indicator variables, *Time*(.), are defined for the eight 6-month periods around January 2014 in the 4-year period, e.g., *Time*(0) equals to 1 if the product was issued during February 2014–July 2014, and *Time*(-1) equals one if the product was issued during August 2013–January 2014, and so forth. Other controls include *Maturity*, *Structure*, *Long Inv Threshold*, *Collateral*, *Infrastructure*, *Securities market*, *ifin institutions*, and *Lothors*. All other variables are defined in the Appendix. Standard errors are clustered at the firm and year levels and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5% and 10% level, respectively.

	Dep. Var. = Product expected yield spread (%)	
	(1)	(2)
IG index	-0.132 (0.105)	-0.302*** (0.0598)
Post default	0.387** (0.155)	
IG index × Post default	-0.241** (0.120)	
IG index × Time(-4)		0.286** (0.0919)
IG index × Time(-3)		-0.0384 (0.0884)
IG index × Time(-2)		0.169 (0.123)
IG index × Time(-1)		-0.0743 (0.0531)
IG index × Time(1)		-0.135*** (0.0364)
IG index × Time(2)		-0.109 (0.0674)
IG index × Time(3)		-0.106* (0.0583)
i.real estate	0.269*** (0.0750)	0.336*** (0.0956)
GDP growth	0.00485 (0.0140)	-0.00715 (0.0111)
Log borrowersize	-0.0270 (0.0171)	-0.0288 (0.0230)
Cons.	YES	YES
Other controls	YES	YES
Time FE	NO	YES
Obs.	7,021	7,021
adj. R-sq	0.317	0.442

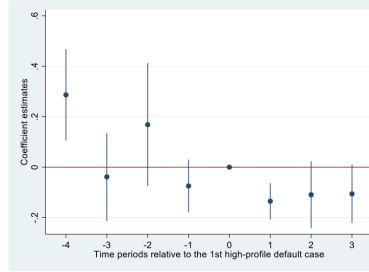


Fig. 4. The first high-profile default case and yield spreads of trust products.

This figure plots the time trend of the treatment effect estimates around the first high-profile trust product default case in January 2014, as explained in Section 5.2 and reported in Column 2 of Table 6. For each time period, we plot the point estimate (the solid circle) and the 95% confidence interval (the vertical lines intersecting the solid circles). Time indicators are defined for the eight 6-month periods around January 2014 in the 4-year period, e.g., *Time*(0) denotes the period during February 2014 – July 2014, and *Time*(-1) denotes the period during August 2013 – January 2014, and so forth.

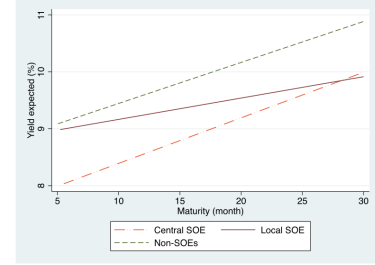


Fig. 5. Yield curve of real estate trust products.

This figure plots predicted yield curves of real estate products issued by different types of trust firms: those ultimately owned by central SOEs, local SOEs, and non-SOEs, respectively. For each group, we regress expected yield on product maturity and calculate the predicted yields. Considering the possible selection issue, we use one-to-one matched product sample based on product features *Log borrowersize* and *GDP growth* (of the location of the project). After matching, we obtain 1,005 products issued by central-SOE-controlled trust companies, 1,005 products issued by local-SOE-controlled trust companies, and 1,005 products issued by non-SOE-controlled trust companies, respectively.

6.9 Fig. 5: yield curve by trust-firm sponsor type

Fig. 5 plots fitted yield curves for real-estate products by trust-firm ownership type. Central SOE-controlled trust firms exhibit the lowest yields across maturities, followed by local SOEs, with non-SOE-related trusts having the highest yields. This is an intuitive “picture proof” of the paper’s pricing mechanism: investors demand lower compensation when they expect stronger implicit support.

6.10 Table 7: real-estate subsample and housing-market risk

Within real-estate trust products, housing-market risk enters positively, and its interaction with IG index is negative and significant (e.g., $\text{IG index} \times \text{Hmarket risk} \approx -0.030$), reinforcing that guarantees reduce risk pricing even within a single sector.

6.11 Table 8 and Fig. 6: housing purchase restrictions as a risk shock

Table 8 uses housing purchase restrictions (Order 10) as a shock that increases real-estate project risk. The RE shock coefficient is large and positive (about 0.627), while $\text{RE shock} \times \text{IG index}$ is

negative (about -0.370), again showing that strong guarantees mitigate the spread response. Fig. 6 provides the event-time visualization of these interaction effects.

Table 7

Real estate product subsample: yield spread and housing market risk. This table reports the results of regressions examining the effect of investor expectation of implicit guarantees on sensitivity of product pricing to housing market risk based on the subsample of real estate products. The dependent variable is the product yield spread. *Hmarket risk* is defined as the residual of the regression of housing prices (adjusted by disposable income per capita) on GDP growth by province (Glaeser et al., 2017). All other variables are defined in the Appendix. Standard errors are clustered at the firm and year levels and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5% and 10% level, respectively.

	Dep. Var. = Product expected yield spread (%)	
	(1)	(2)
IG index	-0.371*** (0.0536)	-1.193** (0.485)
Hmarket risk	0.0162* (0.00973)	0.0606*** (0.0143)
IG index×Hmarket risk		-0.0301*** (0.00770)
IG index×GDP growth		0.000923 (0.00440)
IG index×Log borrowsize		0.0430 (0.0269)
GDP growth		-0.0149* (0.00776)
Log borrowsize		-0.102** (0.0432)
Maturity	0.0416*** (0.0108)	0.0418*** (0.0107)
Structure	0.305*** (0.0883)	0.290*** (0.0903)
Log Inv threshold	0.735*** (0.231)	0.777*** (0.228)
Collateral	0.131 (0.102)	0.0905 (0.0993)
Cons	-2.549** (1.221)	-0.328 (1.202)
Year FE	YES	YES
Province FE	YES	YES
Obs.	3,282	3,282
adj. R-sq	0.466	0.472

Table 8

Restrictions on housing as a negative shock to the underlying risk of real estate products.

This table reports the results of regressions examining the impact of “Order 10” (housing purchase restrictions) in April 2010 on the pricing of implicit guarantee in the real estate industry. The implementation of “Order 10” was staggered across provinces. We include real estate products issued two years around the implementation of the rules. *RE shock* is defined as 1 if the product was issued in the 12 months after the restriction was implemented in each province, and 0 if the product was issued within 12 months before. The indicators *Time(*t*)* are defined eight 3-month periods around the policy shock in the 2-year period, e.g., *Time(0)* equals to 1 if the product was issued in the first three months after the policy shock, and *Time(-1)* equals to 1 if the product was issued in the three months before the policy shock, and so forth. The dependent variable is the product yield spread. Other controls include *Maturity*, *Structure*, *Long Inv Threshold*, *Collateral*, *GDP growth* and year fixed effects. All other variables are defined in the Appendix. Standard errors are clustered at the firm and year levels and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5% and 10% level, respectively.

	Dep. Var. = Product expected yield spread (%)	
	(1)	(2)
IG index	-0.173 (0.109)	-0.0615 (0.154)
RE shock	0.627*** (0.196)	
IG index× RE shock	-0.370*** (0.125)	
IG index× Time(-4)		-0.535 (0.353)
IG index× Time(-3)		-0.0116 (0.208)
IG index× Time(-2)		-0.221 (0.143)
IG index× Time(-1)		0.0779 (0.220)
IG index× Time(1)		-0.716** (0.280)
IG index× Time(2)		-0.494*** (0.184)
IG index× Time(3)		-0.500** (0.237)
Hmarket risk	0.0304 (0.0195)	0.0207 (0.0252)
Cons	YES	YES
Other controls	YES	YES
Time FE	NO	YES
Province FE	YES	YES
Obs	331	331
adj. R-sq	0.185	0.169

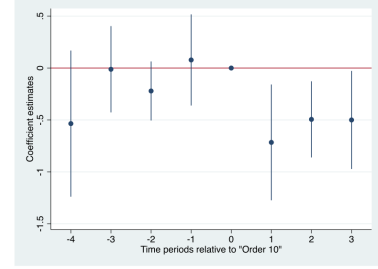


Fig. 6. “Order 10” and yield spreads of trust products.

This figure plots the time trend of the treatment effect estimates around the “Order 10” announced in April 2010, as explained in Section 5.3.1 and reported in Column 2 of Table 8. For each time period, we plot the point estimate (the solid circle) and the 95% confidence interval (the vertical lines intersecting the solid circles). Time indicators are defined eight 3-month periods around the policy shock in the 2-year period, e.g., *Time(0)* denotes the first three months after the policy shock, and *Time(-1)* denotes the three months before the policy shock, and so forth.

6.12 Table 9: matched-sample robustness

Table 9 constructs matched samples of high-IG vs low-IG products to improve comparability on observables (borrower size, GDP growth, industry). Panel A shows balance improvements; Panel B reproduces the key patterns (negative IG effects on spreads; negative interactions with risk shocks), strengthening the causal interpretation.

6.13 Fig. 7: macro context — total social financing

Fig. 7 decomposes China’s total social financing into bank loans, equity/bonds, and entrusted/trust loans. Trust/entrusted loans rise sharply in the 2010s and then decline after 2018, providing macro-level context consistent with a policy-driven contraction in shadow banking.

6.14 Table 10: 2018 regulations weaken the IG–spread relationship

Table 10 estimates the pricing regression around the 2018 asset-management regulations. The interaction *PostReg*×*IGStrength* is positive and significant (e.g., *IG index* × *PostReg* ≈ 0.195), while the *IG index* main effect remains negative. Interpretation:

after regulation, the “spread discount” for high-IG products becomes smaller, consistent with weaker implicit guarantees (or higher cost of providing them).

Table 9

Matched sample tests.
This table reports the results of robustness checks using the propensity score matched (product) sample. The dependent variable is the product yield spread. *Matched* is defined as 1 if the IG index equals 3, 4, or 6, and 0 otherwise. The products with the level of implicit guarantees (non-SOEs) are matched based on log turnover, CDP growth and industry size. A one-to-one propensity score matching algorithm with the distance of 0.01 (logit-Score) is used without replacement. Panel A reports the results of the regression examining the effect of implicit guarantees and product expected yield spreads using the matched sample. Other controls include Minority, Structure, Long for Threshold, Culture, or Culture (1/14) and replacement, turnover, size, IG, industry, and others in Columns (1)–(5). All other variables are defined in the Appendix. Standard errors are clustered at the firm and year levels and reported in the parentheses. ***, **, and * denote statistical significance at the 1%, 5% and 10% level, respectively.

Panel A. Trust product characteristics: matched sample				
	IGIndex=0	IGIndex=1	IGIndex=2	Difference
	OLS	Matched (OLS)	OLS	Matched (OLS)
Log turnover	3.952	18.535	3.952	18.537
	(0.0007)	(0.0012)	(0.0007)	(0.0012)
CDP growth	3.952	18.535	3.952	18.537
	(0.0007)	(0.0012)	(0.0007)	(0.0012)
Panel B. Regression analysis				
	Table 3	Table 5	Table 6	Table 7
	(1)	(2)	(3)	(4)
IG index	-0.427***	-0.756	-0.289	-2.750***
	(0.0002)	(0.0005)	(0.0002)	(0.0002)
CDP growth	-0.0204***	-0.0318***	-0.00227	-0.0174***
	(0.00003)	(0.00003)	(0.0002)	(0.00003)
Local estate	0.211***	0.311***	0.192	0.192
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Log turnover size	-0.0277***	-0.0311***	-0.0288***	-0.0299***
	(0.0004)	(0.0004)	(0.0004)	(0.0004)
IGIndex×CDP growth	0.0197***	0.0197***	0.0197***	0.0197***
	(0.0004)	(0.0004)	(0.0004)	(0.0004)
IGIndex×Local estate	0.0100	0.0100	0.0100	0.0100
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
IGIndex×Log turnover	0.0100	0.0100	0.0100	0.0100
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Post default			0.277	0.277
			(0.0000)	(0.0000)
IGIndex×Post default			-0.0000	-0.0000
			(0.0000)	(0.0000)
IGIndex×Risk			0.0000	0.0000
			(0.0000)	(0.0000)
IGIndex×IGIndex risk			-0.0000	-0.0000
			(0.0000)	(0.0000)
Control variables	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes
Size	7.864	7.864	7.864	7.864
adj. R-sq	0.459	0.442	0.259	0.462

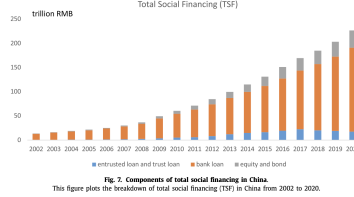


Fig. 7. Components of total social financing in China. This figure plots the breakdown of total social financing (TSF) in China from 2002 to 2020.

Table 10

Policy shock to implicit guarantees: New Regulations on Asset Management in 2018.
This table reports the results of the regression examining the effect of a policy change on implicit guarantees, the “New Regulations on Asset Management” announced in March 2018. The regression is defined as 1 if the product is issued after March 28, 2018, and 0 otherwise. The dependent variable is the product yield spread. Columns (1) and (2) use the full sample and Columns (3) use the propensity score-matched sample, which follows the same matching criteria as shown in Table 9. Other controls include Minority, Structure, Long for Threshold, Culture, or Culture (1/14) and replacement, turnover, size, IG, industry, and others in Columns (1)–(5). All other variables are defined in the Appendix. Standard errors are clustered at the firm and year levels and reported in parentheses.

	(1)	(2)	(3)
	Full sample	Full sample	Matched sample
Control SOE×Post regulation	0.427***	0.427***	0.427***
	(0.0000)	(0.0000)	(0.0000)
Log Reg. size×Post regulation	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)
Size×Log Reg. size×Post regulation	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)
IG index×Post regulation	0.242***	0.199***	0.199***
	(0.0000)	(0.0000)	(0.0000)
Control SOE	-0.262***	-0.262***	-0.262***
	(0.0000)	(0.0000)	(0.0000)
Log Reg. size	-0.179***	-0.179***	-0.179***
	(0.0000)	(0.0000)	(0.0000)
Size×Log Reg. size	-0.0000	-0.0000	-0.0000
	(0.0000)	(0.0000)	(0.0000)
IG index	-0.238***	-0.238***	-0.238***
	(0.0000)	(0.0000)	(0.0000)
Log Reg. size	-0.0000	-0.0000	-0.0000
	(0.0000)	(0.0000)	(0.0000)
CDP growth	-0.0000	-0.0000	-0.0000
	(0.0000)	(0.0000)	(0.0000)
Local estate	0.242***	0.242***	0.242***
	(0.0000)	(0.0000)	(0.0000)
Log turnover size	-0.0000	-0.0000	-0.0000
	(0.0000)	(0.0000)	(0.0000)
IGIndex×Local estate	-0.0000	-0.0000	-0.0000
	(0.0000)	(0.0000)	(0.0000)
IGIndex×Log turnover	-0.0000	-0.0000	-0.0000
	(0.0000)	(0.0000)	(0.0000)
Location×Local estate	-0.0000	-0.0000	-0.0000
	(0.0000)	(0.0000)	(0.0000)
Size×Local estate	-0.0000	-0.0000	-0.0000
	(0.0000)	(0.0000)	(0.0000)
Control variables	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Province FE	Yes	Yes	Yes
Size	7.864	7.864	7.864
adj. R-sq	0.462	0.459	0.467

6.15 Fig. 8 and Table 11: quantity effects and (non-)SOE credit rationing

Fig. 8 shows the share of trust lending to non-SOEs over time and suggests a post-2018 decline. Table 11 formalizes this with issuance-size regressions:

- NonSOE×PostReg is large and negative (credit to non-SOEs contracts after regulation).
- Triple interactions with IG strength (or central SOE status) are positive, implying that stronger-IG intermediaries attenuate the non-SOE contraction.

This is consistent with the model’s “capital misallocation” channel: restricting implicit guarantees reduces credit to riskier private firms and shifts financing back toward SOEs.

7 Short summary

1. **Conceptual:** yields in shadow banking price both borrower risk and *guarantee strength*. Empirically, this means any “credit spread” study in such settings must account for guarantee heterogeneity.
2. **Methodological:** interacting risk shocks with guarantee proxies is a clean way to separate “risk” from “guarantee” in pricing.
3. **Macro/IO angle:** implicit guarantees can be efficiency-enhancing (reduce misallocation) but also create fragility and regulatory tradeoffs. Tightening guarantees can raise measured spreads and simultaneously reduce risky credit supply.

8 Conclusion

8.1 Summary

This paper explains a central feature of China’s shadow banking system: **implicit guarantees** provided by financial intermediaries (trust companies and related institutions) to investors in trust

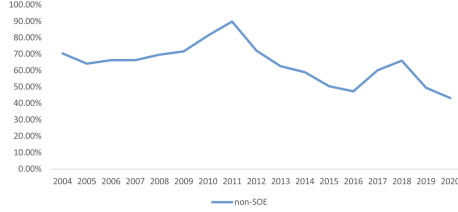


Fig. 8. Percentage of trust loans for non-SOEs. This figure plots the percentage of capital funded by our sample trust products going to non-SOE borrowers from 2004 to 2020. To identify whether the borrower of trust products is an SOE, we match with the registration data originated from the State Administration for Industry and Commerce (SAIC) and China Economic Census Data.

Table 11

The effect on issuance size of trust loans: *New Regulations on Asset Management*.

This table reports the results of the regressions examining the effect of the "New Regulations on Asset Management" announced in March 2018 on the issuance size of the trust loans. *Post regulation* is defined as 1 if the product is issued after March 28, 2018, and 0 otherwise. *Borrower non-SOE* is defined as 1 if the underlying borrower is a non-SOE, and 0 otherwise. The dependent variable is the natural logarithm of the issuance size of trust loans. The sample covers the product lending to real estate and Commercial and Industrial (C&I) firms. Other controls include *Maturity*, *Structure*, *Long Inv Threshold*, *Collateral*. All other variables are defined in the Appendix. Standard errors are clustered at the firm and year levels and reported in the parentheses.

	Dep. Var.=Log Loan size	
	(1)	(2)
Borrower non-SOE×Post regulation	-1.548*** (0.471)	-2.327** (0.904)
Central SOE× Borrower non-SOE×Post regulation	2.101*** (0.712)	
IG index × Borrower non-SOE×Post regulation		1.073** (0.425)
Central SOE×Borrower non-SOE	0.196 (0.274)	
Central SOE×Post regulation	0.109 (0.793)	
Log Reg cap×Post regulation	-0.00222 (0.108)	
Sale bank big 5×Post regulation	2.183*** (0.753)	
IG index×Post regulation		0.0292 (0.195)
IG index ×Borrower non-SOE		-0.0932 (0.124)
Central SOE	-0.0400 (0.311)	
Log Reg cap	0.0563 (0.146)	
Sale bank big 5	-0.488*** (0.182)	
IG index		0.0906 (0.100)
Borrower non-SOE	0.0486 (0.105)	0.158 (0.194)
Log Shrd size	0.0593 (0.0494)	0.0561 (0.0443)
GDP growth	0.0240 (0.0210)	0.0339 (0.0226)
i.real estate	0.167* (0.0881)	0.222** (0.0859)
Log borrower size	0.169*** (0.0282)	0.186*** (0.0397)
Cons/Other controls	YES	YES
Year FE	YES	YES
Province FE	YES	YES
Obs	1,100	1,100
adj. R-sq	0.229	0.216

products. Although prospectuses state that returns are not contractually guaranteed, investors commonly believe that intermediaries (and sometimes affiliated state entities) will cover losses in typical states of the world.

The paper makes two linked contributions:

1. **Theory.** In a delegated-monitoring environment where investors rely on an intermediary to screen projects, an *implicit* (state-contingent) guarantee can be **second-best optimal**. It creates *skin in the game* for the intermediary (thus incentivizing screening) while allowing non-payment in rare aggregate bad states when the cost of raising funds to honor the guarantee is exceptionally high.
2. **Evidence.** Using a comprehensive dataset of trust products, the paper shows that **ex ante yield spreads price both (i) underlying borrower/project risk and (ii) heterogeneity in implicit guarantee strength**. Stronger guarantees lower spreads and dampen the sensitivity of spreads to risk. After the 2018 asset-management regulations that restricted implicit guarantees, the yield-guarantee relationship weakens and **credit to non-SOE borrowers contracts**, consistent with a worsening of capital allocation toward private firms.

8.2 Main conclusions

1. **Spreads price guarantee strength, not only borrower risk.** In the baseline pricing regressions (Table 3, Panel B), the IG index is strongly negatively related to the promised yield spread:

$$\hat{\beta}_{IG} \approx -0.354 \quad (\text{s.e.} \approx 0.040),$$

i.e., moving one unit up in the implicit-guarantee index reduces spreads by about 0.35 percentage points, holding risk and product characteristics fixed.

2. **Investors react to shocks that weaken intermediaries' guarantee capacity.** Using the 2015 stock market crash as a shock to trust-firm balance sheets, the interaction between sponsor stock exposure and the post-crash indicator is positive (Table 4):

$$\hat{\beta}_{\text{StkExposure} \times \text{PostCrash}} \approx 0.0719 \quad (\text{s.e.} \approx 0.0348),$$

consistent with higher spreads when the sponsor's perceived ability to backstop losses deteriorates.

3. **Guarantees reduce risk pricing (the spread-to-risk slope).** Interaction tests show that stronger guarantees dampen how strongly spreads respond to underlying risk proxies (Table 5 and related event-shock designs), consistent with the model's prediction that guarantees partially absorb downside risk in normal states.
4. **Policy tradeoff: restricting implicit guarantees weakens the pricing role of guarantees and tightens credit to non-SOEs.** After the "New Regulations on Asset Management" (March 28, 2018), the spread advantage of high-IG products shrinks. In Table 10,

$$\hat{\beta}_{IG \times \text{PostReg}} \approx 0.195 \quad (\text{s.e.} \approx 0.086),$$

meaning the negative IG-spread slope becomes materially weaker post-regulation (investors perceive less protection, or guarantees become costlier to provide).

5. **Real effect: non-SOE credit rationing rises after 2018, especially when guarantees are weak.** In issuance-size regressions (Table 11), lending to non-SOE borrowers contracts sharply after regulation:

$$\hat{\beta}_{\text{NonSOE} \times \text{PostReg}} \approx -1.548 \quad (\text{s.e.} \approx 0.471),$$

but the contraction is *attenuated* by stronger guarantee capacity:

$$\hat{\beta}_{IG \times \text{NonSOE} \times \text{PostReg}} \approx 1.073 \quad (\text{s.e.} \approx 0.425), \quad \hat{\beta}_{\text{CentralSOE} \times \text{NonSOE} \times \text{PostReg}} \approx 2.101 \quad (\text{s.e.} \approx 0.712).$$

This pattern matches the paper's misallocation logic: removing/penalizing guarantees disproportionately reduces funding to private (non-SOE) borrowers.

8.3 Economic intuition: why the results fit together

- **Delegated screening needs commitment.** Investors cannot (or do not) screen underlying projects. If intermediaries can originate risky loans without bearing losses, screening incentives are weak.

- **Implicit guarantees create intermediary skin in the game.** By (implicitly) committing to cover losses in normal times, intermediaries internalize downside risk and are incentivized to screen and monitor.
- **Why “implicit” (state-contingent) rather than explicit?** Honoring guarantees requires raising liquidity. In aggregate bad states, external finance is exceptionally costly, so the efficient second-best contract allows guarantees to fail in those states (this preserves feasibility and avoids extreme deadweight costs).
- **Why this matters for China specifically (misallocation).** Traditional finance in China has historically favored SOEs. Shadow banking with implicit guarantees expands credit access for non-SOEs. When regulations restrict guarantees, risk-bearing shifts back to investors and/or intermediaries cannot credibly backstop losses, so funding to riskier private borrowers contracts, raising capital misallocation.

8.4 Takeaway

Implicit guarantees in trust products are not just “distortions”: they can be a second-best tool that sustains screening and reallocates credit toward non-SOEs, but tightening them (as in 2018) weakens guarantee pricing and increases credit rationing of private borrowers.